



*Software Requirements Specification
Version 1.0*

AI Attendance Manager

Theme: Smart Attendance System Using Face Recognition

Project Name: AI Attendance Manager

Category: AI and Machine Learning Mania

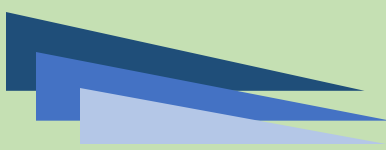


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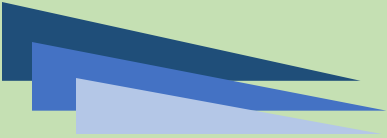
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1.1 Background and Necessity for the Application

Attendance management is vital in educational institutions, directly impacting student's performance and academic outcomes. Traditional methods required manual attendance, that was time-consuming and prone to errors such as proxy marking and difficulties with large class sizes. The **AI Attendance Manager** aims to automate this process, reducing faculty workload, improving accuracy, and preventing proxy attendance through smart and AI-driven solutions.

'**AI Attendance Manager**' eliminates the requirement of manual attendance preventing proxy attendance. This application addresses the requirement by automatically recording students' attendance through facial recognition as they enter the classroom. It also tracks In-time and Out-time, allowing accurate calculation of each student's presence duration. By relying on facial identity, the system ensures authenticity and eliminates the possibility of fake attendance. The goal is to deliver a reliable, secure, and accessible solution that enhances overall classroom management.

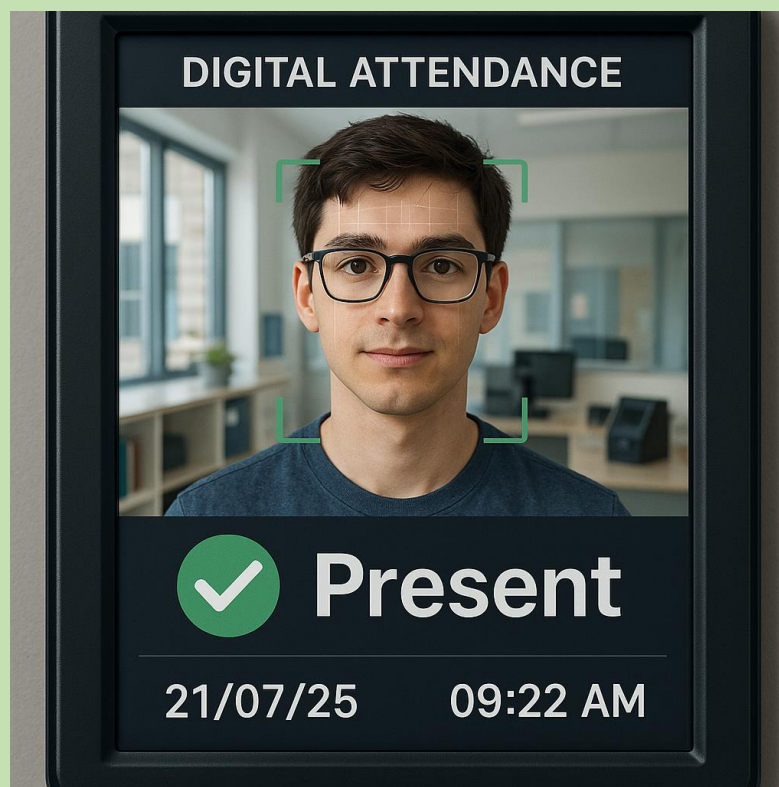




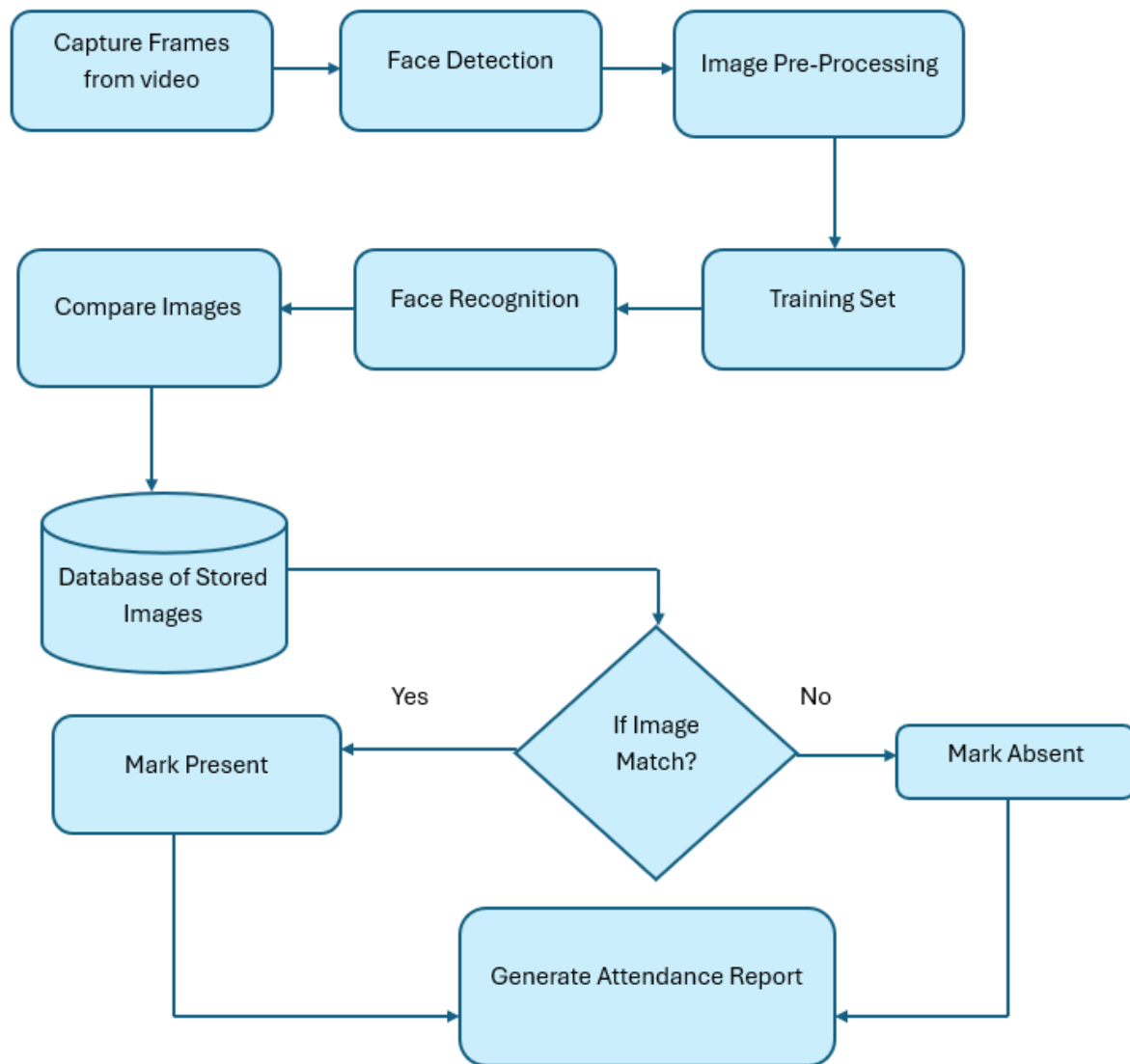
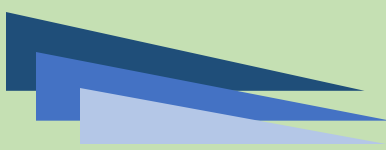
1.2 Proposed Solution

The proposed solution introduces an advanced, AI-powered attendance system that leverages facial recognition technology to automate and secure the attendance process. By using a camera module installed in the classroom, the system captures images of students as they enter. These images are then, matched against a pre-stored database using facial recognition algorithms. When a match is found, the system automatically logs the student's attendance, eliminating the necessity for manual marking and significantly reducing the chance of errors.

AI Attendance Manager accurately marks attendance by verifying that only students who are physically present in the classroom are recorded as present, eliminating the risk of proxy attendance. Its implementation promises enhanced accuracy, real-time tracking, and a streamlined process that saves time for faculty and administrative staff. By automating attendance and improving data reliability, the system addresses common challenges in traditional methods and supports smarter and more efficient academic environment.



The sample architecture for AI Attendance Manager application is as follows:

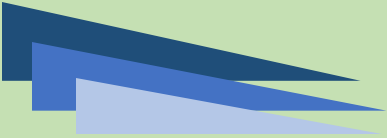


Sample Architecture of the Application

The development phase of the application includes following steps:

Step 1: Capturing Video Frames

A camera is positioned at the classroom entrance to capture clear images of students' faces. It records video at 30 frames per second. These frames are extracted and used for further processing.



Step 2: Face Detection

This phase includes face detection of the person which aids in locating and sizing the student faces in the collected image.

Step 3: Image Pre-Processing

Pre-processing is necessary to enhance the input image and raise the overall quality of the image.

Step 4: Training Set

The method involves comparing the faces that has to be recognized with the identical face in the trained dataset. The provided algorithm uses the training dataset to identify which group a person belongs to. It relies on this training data to perform facial recognition and make accurate classifications.

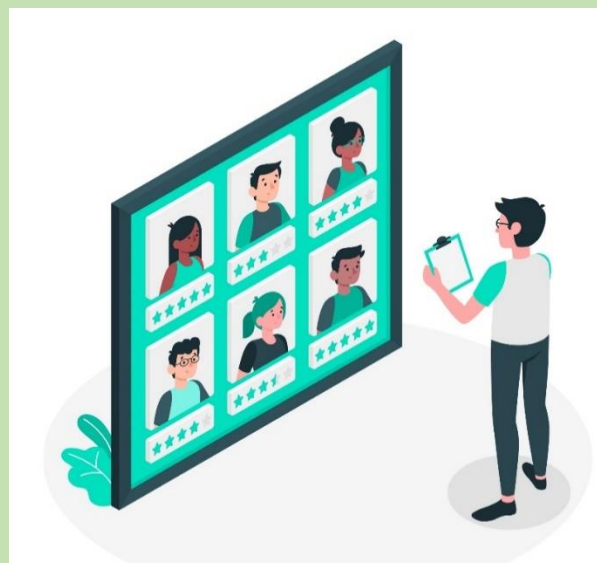
Step 5: Face Recognition

The face recognition component plays a crucial role in the system. An automated way of recognizing and authenticating a person from photos and videos taken by the camera is face recognition.

Step 6: Marking Attendance

Attendance will only be recorded as present if a face from a particular registered folder matches. Once all present students have been marked, the remaining enrolled students who were not detected will be automatically recorded as absent.

Hint: For this project, you are expected to create your own dataset using real-time image data. For instance, the system can be trained on six different students by capturing over 100 images per student. These images should be converted to grayscale and saved with the respective student's name and ID. Please note that no pre-existing dataset is provided for this task.

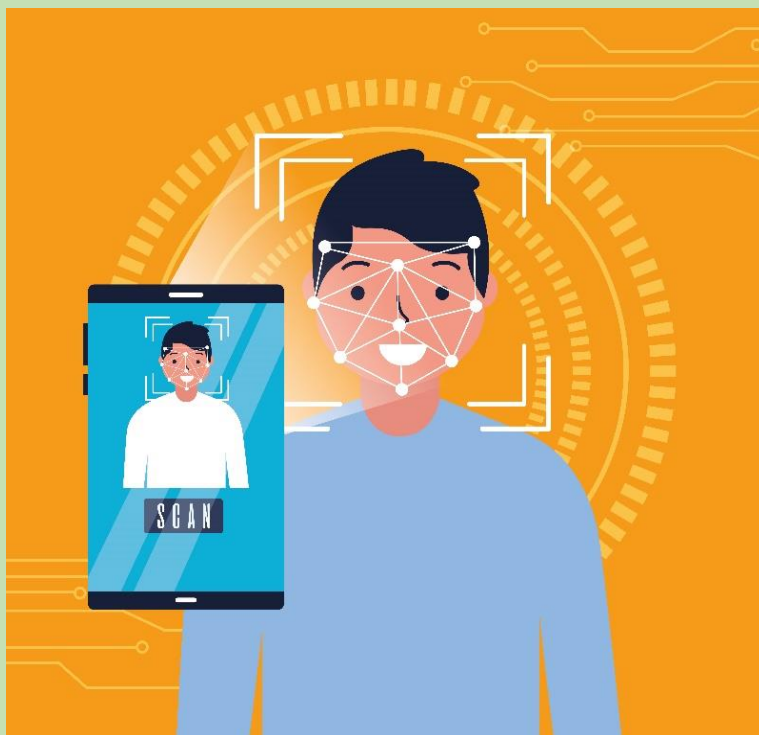


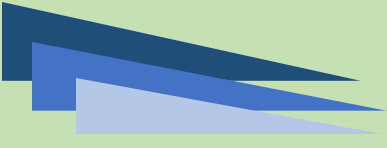
1.3 Purpose of the Document

The purpose of this document is to outline the design, functionality, and implementation plan of the proposed Smart Attendance System Using Facial Recognition. This document ensures a shared understanding of the project goals and provides a structured roadmap for development and deployment. This document is intended for both stakeholders and developers of the application.

1.4 Scope of Project

The scope of this project is to design and implement an AI-powered attendance management system that automates student attendance using facial recognition technology. The system will capture student images at the classroom entrance, compare them with a stored database, and mark attendance in real time. It aims to eliminate proxy attendance, reduce manual effort, ensure accuracy, and generate reliable attendance reports. Furthermore, it should be noted that support integration with existing academic systems falls outside the scope of this project.





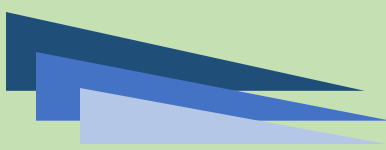
1.5 Constraints

The AI Attendance Manager project faces certain constraints such as dependency on good lighting conditions and proper camera placement for accurate facial recognition. The system's effectiveness is also limited by the quality and size of the training dataset. Privacy concerns regarding the storage and handling of biometric data must also be addressed in accordance with data protection regulations.

1.6 Functional Requirements

Creating an AI Attendance Manager involves several steps and components. Here are the functional requirements to develop such a system:

- i. Data Collection** - The system should gather and store student facial data as part of their digital profiles to enable accurate facial recognition for automated attendance tracking.
- ii. Face Detection** - The system should capture real-time images using a camera at the classroom entrance and detect and recognize student faces.
- **iii. Face Recognition** - Implement face recognition authentication for identification.
- iv. Data Pre-processing** - Clean and pre-process the data to improve the accuracy of the recognition system. This may involve image enhancement, resizing, and data normalization.
- v. Machine Learning Model** - Train a Machine Learning Model (example, Convolutional Neural Network (CNN)) to recognize faces. You may use pre-trained models and fine-tune them for your specific dataset.
- **vi. Student Identification** - The system should match the captured face with stored images in the student database and identify the student using unique identifiers such as name or ID.
- vii. Attendance Marking** - If a match is found, the system should automatically mark the student as present. If not, the student is marked as absent.



viii. Time Logging - The system should record the in-time and out-time of each student to monitor the total time spent in class.



ix. Data Storage - Set up a database to store attendance records securely. Consider using relational databases or NoSQL databases, depending on your scalability requirements.

x. User Interface - Develop a user-friendly interface for users to mark attendance, view attendance records, and manage the system. This can be a Web based application.

xi. Admin dashboard - The system should provide a dashboard for faculty/admins to view attendance reports, export data, and manage student profiles.

xii. Proxy Detection prevention - The system should ensure each face is unique and prevent multiple entries or proxy attendance.

xiii. Data Analysis and Reporting - Implement data analytics to generate reports on attendance patterns, absenteeism, and other relevant metrics.

In addition, the system should set up monitoring tools to detect anomalies or issues with the system and configure alerts for timely responses. Also, it should send alerts for irregular attendance if required.

1.7 Non-Functional Requirements

There are several non-functional requirements that should be fulfilled by the application which are as follows:



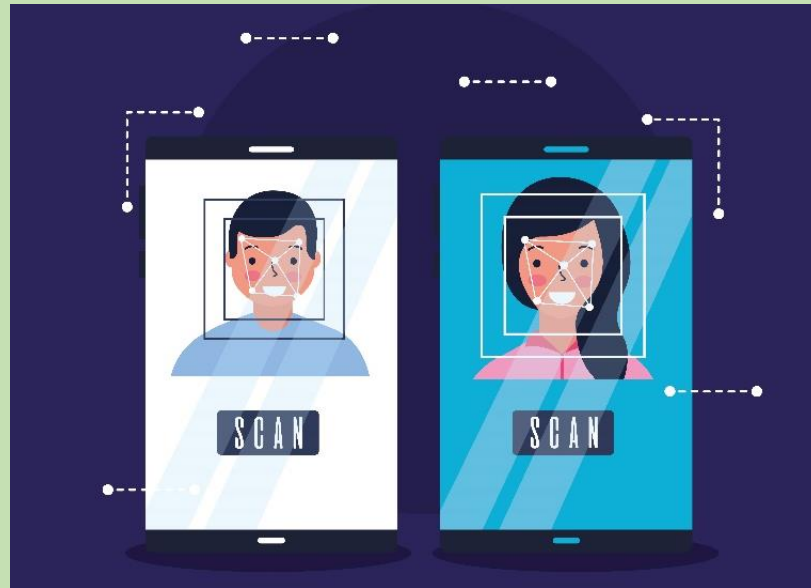
1. Performance: The system should process and recognize a student's face within two seconds to ensure minimal delay during attendance marking.

2. Scalable: The system should be able to handle multiple classrooms or lecture halls and recognize up to 1000 students without degradation in performance.

3. Usable: The system should have an intuitive and user-friendly interface for administrators, teachers, and students.

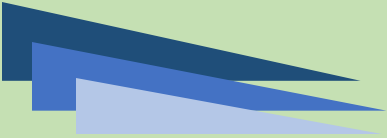


4. **Accuracy**: The facial recognition algorithm should maintain an accuracy rate of at least 95% to ensure reliable attendance records.
5. **Available**: The system should be available and operational during all working hours with 99% uptime to ensure continuous attendance tracking.



These are the bare minimum expectations from the project. It is a must to implement the **FUNCTIONAL** and **NON-FUNCTIONAL** requirements given in this SRS.

Once they are complete, you can use your own creativity and imagination to add more features if required.



1.8 Interface Requirements

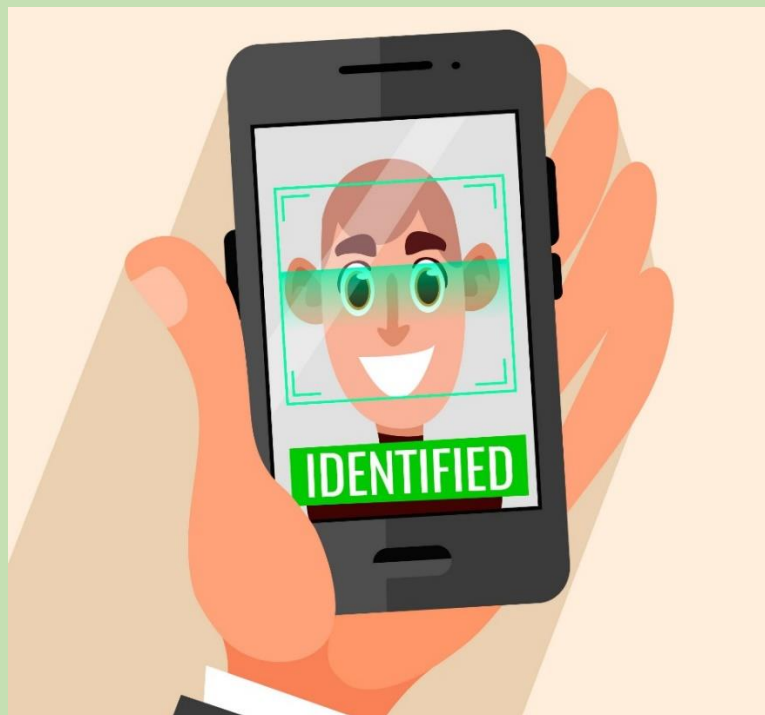
1.8.1 Hardware

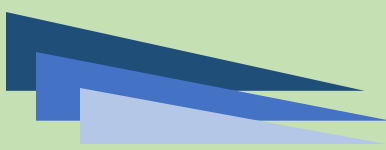
Intel Core i5/i7 Processor or higher
8 GB RAM or higher
Color SVGA monitor
500 GB Hard Disk space
Mouse and Keyboard

1.8.2 Software

Technologies to be used:

1. **Frontend:** HTML5/Streamlit or any other frontend programming languages
2. **Backend:** Flask/Django
3. **Data Store:** jpg images
4. **Database:** Firebase/NoSQL
5. **Programming/IDE:** Python, Jupyter Notebook, Anaconda, Google Colab
6. **Libraries:** OpenCV, NumPy, Pandas, Matplotlib, Keras, Tensorflow, PyTorch





1.9 Project Deliverables

You will design and build the project and submit it along with a complete project report that includes:

- Problem Definition
- Design specifications
- Diagrams such as Dialog Flow
- Test Data Used in the Project
- Project Installation Instructions
- Proper Steps to execute the project
- Link of GitHub for accessing the uploaded project code (Link should have public access)

The source code, including .ipynb files for Jupyter Notebook and Google Colab, should be shared via GitHub. Appropriate access permissions should be granted to users to allow testing for Jupyter Notebook and Google Colab. The consolidated project must be submitted on GitHub with a ReadMe.doc file listing assumptions (if any) made at your end.

Please provide the GitHub URL where the project has been uploaded for sharing. The repository on GitHub should have public access. Documentation is a very important part of the project; hence, all crucial aspects of the project must be documented properly. Ensure that documentation is complete and comprehensive.

You should publish a blog of minimum 2000 words on any free blogging Website such as Blogger, Tumblr, Ghost, or any other blogging Website. The link of the published blog should be submitted along with the project documentation.

Do NOT copy content or code from GPTs or other AI tools, although you are permitted to use images generated by AI tools for any visual representation purposes. It is mandatory to mention such tools used in case you add any AI generated images.

Submit a video (.mp4 file) demonstrating the working of the application, including all the functionalities of the project. This is MANDATORY.

~~~ *End of Document* ~~~